Zhuoer Du

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Education

University of Michigan, College of Engineering Ann Arbor, MI

Bachelor of Science in Engineering in Computer Science, Minor in Statistics *Aug. 2024 – May 2028*

- **Honors/Awards:** UM Club of Houston Scholarship, Dean's Honor List, University Honors
- **Relevant Coursework:** Data Structures & Algorithms, Discrete Math, Computer Organization, Web Systems, Statistical Computing, Computer Security, Computer Networks

Experience

Taubman Health Sciences Library Jan 2025 – Present

Information Desk Assistant *Ann Arbor, MI*

- Supported 15-25 patrons per shift by identifying needs and resolving navigation, access, and space-reservation issues efficiently.
- Managed circulation operations for 200-300 materials per week, maintaining accurate records and reliable access to shared resources.
- Processed and organized 20-30 library holds and reserves daily, improving turnaround time and supporting timely access to high-demand materials.

Software Engineer Sept. 2024 – Present

Tech 4 Social Good *Ann Arbor, MI*

- Developed a Python-based data processing pipeline to transform raw drainage and infrastructure datasets into GIS-compatible formats, enabling adoption by 4 county offices.
- Automated manual data-cleaning and conversion workflows, reducing processing time by approximately 60% and minimizing human error in recurring reports.
- Collaborated with county stakeholders and a cross-functional student team to translate real-world government requirements into production-ready tooling

Data Scientist Jan. 2025 – Present

Michigan Data Science Team *Ann Arbor, MI*

- Trained a CNN-based image classification model using TensorFlow to identify fruit types from images, integrating predictions with a nutrition API to deliver real-time nutritional information through a user-facing application.
- Cleaned, labeled, and processed 1000+ images from public datasets, improving model accuracy to approximately 70% through iterative training and evaluation.
- Contributed to an AlphaFold-inspired protein structure prediction project, analyzing biological sequence data and exploring deep learning architectures for modeling protein folding behavior.

Lab Mentor Aug. 2023 – May 2024

iCode Cypress *Cypress, TX*

- Taught 60+ students (ages 8-18) foundational programming concepts through hands-on lessons in HTML, CSS, JavaScript, and Unity, adapting instructions to different learning levels.
- Designed and delivered interactive coding projects and lesson plans, improving student engagement and enabling learners to build functional games and web applications.
- Provided individualized guidance and debugging support, helping students complete projects successfully and build confidence in problem-solving and logical thinking.

Projects

MapReduce | *Python, Distributed Systems, Parallel Processing, Networking, Process Management* December 2025

- Built a backend processing system that splits jobs into smaller tasks and runs them in parallel across 2+ worker processes, improving execution efficiency.
- Implemented job coordination and failure handling, automatically reassigning work after missed health checks to ensure 100% job completion during testing.
- Processed 8+ input files/records per job, validating correctness and performance through automated tests and simulated failure scenarios.

Insta485 | *React, JavaScript, Flask, REST API, SQL, AWS, Webpack, HTML/CSS* November 2025

- Developed a React-based social media feed that consumes REST APIs to render content client-side, implementing infinite scroll with batched loading of 10 posts per request for a smooth user experience.
- Designed and integrated RESTful backend endpoints for posts, likes, and comments, enabling real-time interactions with authenticated access control and robust error handling.
- Implemented interactive UI features including like/unlike, double-click to like, and live comment updates—and built auto-refreshing timestamps to keep content current without page reloads.

Leadership Experience

Event Logistics & Organizations Committee Member Jan. 2026 – Present

Girls in Electrical Engineering and Computer Science (GEECS) *Ann Arbor, MI*

- Coordinate logistics for 3 community events per month, managing room reservations, materials, and day-of execution to support programming for 20+ attendees per event.
- Support mentorship program of 30+ students by organizing monthly family meetings, tracking engagement, and facilitating activities that strengthen peer connections and participation.
- Work with a cross-functional committee to develop programming ideas and organize workshops, socials, and speaker events, contributing to initiatives that maintain strong member engagement across the organization.

Product Manager Jan. 2026 – Present

MProduct *Ann Arbor, MI*

- Lead product development for a housing-focused application, conducting user interviews to identify common pain points and synthesizing insights to define core product requirements and feature priorities.
- Design the product experience by creating wire frames, user flows, and interactive prototypes, translating user needs into structured layouts and visual concepts that guide the application's functionality and usability.
- Develop high-fidelity interface designs in Figma, iterating on prototypes based on team feedback and user insights while collaborating with teammates to refine the product's overall design and feature set.

Technical Skills

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, R
Frameworks: React, Node.js, Flask
Developer Tools: Git, Docker, VS Code, PyCharm, Eclipse
Libraries: pandas, NumPy, Matplotlib